BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: MRD-XII-21 (A) [INCOMPLETE: Subjective not found]

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** MRD-XII-21 (A) VERSION: A

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading. [INCOMPLETE PAPER: Subjective Sections B & C not found on any source — objective only.]

Q1. Choose the correct option i.e. A,B,C, or D.

Q1. When a dielectric medium is inserted between the two-point charges then the magnitude of electrostatic force ....

(A) Increases
(B) Decreases
(C) Becomes zero
(D) Not affected

Q2. Electric field intensity due to an infinite sheet of charge is given by .....

(A) $\sigma/2\varepsilon_0$
(B) σ/ε_0
(C) $\sigma/\pi\varepsilon_0$
(D) σ/q

Q3. In Ohm's Law R = V/I, so by doubling the voltage, the resistance will ....

(A) Double
(B) Half
(C) Four-times
(D) Remains the same

Q4. Which of the following is an example of wire-wound variable resistors?

(A) Potentio meter
(B) Thermistors
(C) Rheostats
(D) Wheat stone bridge

Q5. Ampere's circuital law is given by the relation .....

(A) $B = \mu_0 I$
(B) $B = \mu_0 I/r$
(C) $B = \mu_0 I/2\pi r$
(D) $B = \mu_0 Ir$

Q6. One weber is equal to .....

(A) $N \cdot A^2/m$ (B) $N \cdot m^2/A$

(C) $N \cdot A/m$ (D) $N \cdot m/A$

Q7. Which of the following quantities remain constant in step-up transformer?

(A) Power (B) Current

(C) Voltage (D) Heat

Q8. What is the coefficient of mutual inductance, when the magnetic flux changes by 2×10^{-2} Wb, and change in current is 0.01A?

(A) 1/2 H (B) 2H

(C) 3H (D) Zero

Q9. In pure capacitive circuit the power loss is .....

(A) Maximum (B) Minimum

(C) Infinite (D) Zero

Q10. The peak value of alternating current is $5\sqrt{2}$ A. The mean square value of current will be .....

(A) 5A (B) 2.5A

(C) $5\sqrt{2}$ A (D) 5^2 A

Q11. Which of the following is paramagnetic material?

(A) Copper (B) Antimony

(C) Zinc (D) Bismuth

Q12. In transistor, there are ..... PN junctions.

(A) Four (B) Three

(C) Two (D) Five

Q13. The laws of physics are the same in all inertial reference frames. It is stated by .....

(A) Bohr (B) Thomson

(C) Einstein (D) Galileo

Q14. The maximum K.E of emitted photoelectron depends upon .....

(A) Frequency of the incident light (B) Temperature of the metal surface

(C) Intensity of the incident light (D) None of these

Q15. According to Bohr's theory of hydrogen atom the radius of the first orbit is .....

(A) 13.6×10^{-9} m (B) 3.4×10^{-10} m

(C) 5.3×10^{-9} m (D) 0.53×10^{-10} m

Q16. In Bohr theory, the energy of an electron in 3rd quantized orbit is .....

(A) -3.4 eV
(B) −1.51 eV
(C) −0.85 eV
(D) −13.6 eV

Q17. In terms of energy 1 a.m.u = ...

(A) 0.51 MeV
(B) 931 MeV
(C) 937 MeV
(D) 938 Me

Q18. A radioactive substance has a half-life of four months. 3-fourth of the substance will decay in .....

(A) 16 months
(B) 12 months
(C) 8 months
(D) 6 months